

Combinatorics – 7

Extremal set systems and additive combinatorics

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Theoretical minimum. How large can a graph or a family of sets be without a given configuration, and how small can a sumset $A + B$ be? The answers come from double counting with convexity, from induction on the ground set, and from splitting by binary digits. Sperner, Turán, Ramsey and Erdős–Ko–Rado are in Combinatorics – 5 and are used freely.

Definitions. $K_{s,t}$ is the complete bipartite graph with parts of sizes s and t . The *shift graph* G_n has as vertices the pairs (i, j) , $1 \leq i < j \leq n$, and joins (i, j) with (j, k) whenever $i < j < k$. A family $\mathcal{F} \subseteq 2^{[n]}$ is an *up-set* if $A \in \mathcal{F}$, $A \subseteq B$ imply $B \in \mathcal{F}$ (a *down-set* dually), *intersecting* if any two members meet, and *union-closed* if $A, B \in \mathcal{F} \Rightarrow A \cup B \in \mathcal{F}$. In an abelian group $A + B = \{a + b : a \in A, b \in B\}$ and $hA = A + \dots + A$ (h times).

Theorem 1 (Kővári–Sós–Turán). Let G be bipartite with parts U, W , $|U| = m$, $|W| = n$, and suppose no s vertices of U have t common neighbours. Then

$$e(G) \leq (t-1)^{1/s} m n^{1-1/s} + (s-1)n.$$

Proof. Count pairs (S, w) with $S \subseteq U$, $|S| = s$, $S \subseteq N(w)$: $\sum_{w \in W} \binom{d(w)}{s} \leq (t-1) \binom{m}{s}$. The function $x \mapsto \binom{x}{s}$ (set to 0 for $x < s-1$) is convex, so Jensen gives $n \binom{e/n}{s} \leq (t-1) \binom{m}{s}$, and $\binom{x}{s} \geq (x-s+1)^s/s!$ finishes. Consequently a $K_{s,t}$ -free graph on n vertices ($s \leq t$) has at most $\frac{1}{2}(t-1)^{1/s} n^{2-1/s} + \frac{1}{2}(s-1)n$ edges.

Theorem 2 (shift graphs). $\chi(G_n) = \lceil \log_2 n \rceil$, although G_n has no triangle. *Proof.* Colour (i, j) by the highest binary digit in which $i-1$ and $j-1$ differ: there j has a 1, while in the colour of (j, k) it is j that has a 0. Conversely, given a proper colouring c with m colours put $S_i = \{c(i, j) : j > i\}$; for $i < j$ the colour $c(i, j)$ lies in S_i but not in S_j , so the n sets $S_i \subseteq [m]$ are distinct and $n \leq 2^m$.

Theorem 3 (Harris–Kleitman). If $\mathcal{A}, \mathcal{B} \subseteq 2^{[n]}$ are up-sets, then $|\mathcal{A} \cap \mathcal{B}| 2^n \geq |\mathcal{A}| |\mathcal{B}|$; the same holds for two down-sets, and the inequality reverses

for an up-set and a down-set. *Proof* by induction on n : split $\mathcal{A}$ into $\mathcal{A}_0 = \{A : n \notin A\}$ and $\mathcal{A}_1 = \{A \setminus \{n\} : n \in A\}$; then $\mathcal{A}_0 \subseteq \mathcal{A}_1$, and Chebyshev's inequality $2(a_0b_0 + a_1b_1) \geq (a_0 + a_1)(b_0 + b_1)$ for $a_0 \leq a_1, b_0 \leq b_1$ closes the induction.

Theorem 4 (Cauchy–Davenport; subgroups). For non-empty $A, B \subseteq \mathbb{Z}/p\mathbb{Z}$, p prime, $|A + B| \geq \min\{p, |A| + |B| - 1\}$; in $\mathbb{Z}$ always $|A + B| \geq |A| + |B| - 1$ (the proof by the Combinatorial Nullstellensatz is Linear algebra – 6, homework 10). In a finite abelian group G : if $|A| + |B| > |G|$ then $A + B = G$ (pigeonhole), and $|A + A| = |A|$ exactly when A is a coset of a subgroup. In $\mathbb{F}_2^n$ one has $A + A = A - A$, and the extremal sets are affine subspaces.

Fact 1 (covering pairs; designs). If b blocks of size k cover all pairs of a v -set, every point lies in at least $\lceil \frac{v-1}{k-1} \rceil$ blocks, so $b \geq \lceil \frac{v}{k} \lceil \frac{v-1}{k-1} \rceil \rceil$ (Schönheim). A Steiner triple system (every pair in exactly one triple) needs $v \equiv 1, 3 \pmod{6}$; the lines $\{x, y, x + y\}$ of $\mathbb{F}_2^m \setminus \{0\}$ give one for $v = 2^m - 1$ (the Fano plane for $m = 3$).

Fact 2 (union-closed sets; dyadic arguments). Frankl's conjecture: a union-closed family with a non-empty member has an element lying in at least half of the members. It is open; some element always lies in a fraction ≈ 0.38 of them (Gilmer and successors). If a set of integers contains, with any two elements of the same parity, their midpoint, and contains a and $a + 2^k d$, then it contains every $a + jd$, $0 \leq j \leq 2^k$: halve k times. The 2-adic valuation of the gap decides whether such an argument can work.

1. Prove Theorem 1 and deduce the bound for $K_{s,t}$ -free graphs on n vertices; check that for $s = t = 2$ it gives the four-cycle bound of Combinatorics – 5. Then prove that every $n \times n$ matrix of zeros and ones with at least $2n^{5/3}$ ones contains a 3×3 submatrix (three rows, three columns, not necessarily adjacent) consisting of ones, for all n .
2. Let A, B be non-empty subsets of a finite abelian group G . Prove that $|A| + |B| > |G|$ implies $A + B = G$, and that $|A + A| = |A|$ holds exactly for the cosets of subgroups. Then prove that if $|A - A| < \frac{3}{2}|A|$, then $A - A$ is a subgroup (show first that $|A \cap (A + x)| > \frac{1}{2}|A|$ for every $x \in A - A$), and show that the constant $\frac{3}{2}$ cannot be increased.

3. There are $2n$ students in a school ($n \in \mathbb{N}$, $n \geq 2$). Each week n students go on a trip. After several trips the following condition was fulfilled: every two students were together on at least one trip. What is the minimum number of trips needed for this to happen? (IMC 2013)
4. Let $n \geq k$ be positive integers, and let $\mathcal{F}$ be a family of finite sets with the following properties: (i) $\mathcal{F}$ contains at least $\binom{n}{k} + 1$ distinct sets containing exactly k elements; (ii) for any two sets $A, B \in \mathcal{F}$, their union $A \cup B$ also belongs to $\mathcal{F}$. Prove that $\mathcal{F}$ contains at least three sets with at least n elements. (IMC 2016)
5. Let $n > 3$ be an integer. Let Ω be the set of all triples of distinct elements of $\{1, 2, \dots, n\}$. Let m denote the minimal number of colours which suffice to colour Ω so that whenever $1 \leq a < b < c < d \leq n$, the triples $\{a, b, c\}$ and $\{b, c, d\}$ have different colours. Prove that

$$\frac{1}{100} \log \log n \leq m \leq 100 \log \log n.$$

(IMC 2022)